GERARD

Firmware Gospel

Firmware v.1.1.20261111T012239Z: what it does, pins, settings and faults

Release	Version 1.1 (tag `v.1.1`), declared 2026-11-10
Hardware revision	A
Firmware version	`v.1.1.20261111T012239Z`
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Author	Zeroshin

GERARD is a desk focus timer. It counts focus time, counts down a set time, and reminds the person at the desk to take breaks, drink and stretch.

Colophon

Item	Value
Release	Version 1.1, declared 2026-11-10
Exported	2026-11-10
Hardware revision	A
Firmware version	`v.1.1.20261111T012239Z`
Author	Zeroshin
Licences	This document: `CC-BY-SA-4.0`. Hardware design (KiCad and STL files): `CERN-OHL-S-2.0`. Firmware: `GPL-3.0-or-later`.
Root repository commit	`2844412`

Callouts in this document:

> NOTICE. Risk of damage to GERARD or to what it is plugged into, and how to avoid it.

> Note. A fact not to miss. It is not a hazard.

The other book, the Markdown editions of both books and every source file are in the GERARD repository: start with its `README.md`.

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1 The firmware

GERARD's firmware runs on the RP2040-Zero board. It is written for the Arduino core of the RP2040 and built with PlatformIO. Its source is in `firmware/`, and every pin it uses is assigned in one file, `firmware/src/pins.h`. This document describes the firmware to the best of its authors' knowledge.

GERARD is worked with one knob that is also a push button. Three actions exist: turn, press, and hold (a press held 0.7 s or more by default; Table 5). The knob moves one step per click. With sound on, each click ticks and each press clicks.

2 Pins

Table 1. Pins, as `firmware/src/pins.h` assigns them.

Signal	Pad	Direction	Connected to
`ENC_CLK`	GP0	Input, pull-up	Encoder, pin CLK
`ENC_DT`	GP1	Input, pull-up	Encoder, pin DT
`ENC_SW`	GP2	Input, pull-up; low when pressed	Encoder's push button, pin SW
`I2C_SDA`	GP4	I2C0 data	Display, pin SDA
`I2C_SCL`	GP5	I2C0 clock, 400 kHz	Display, pin SCK
`BUZZER`	GP8	Output, high for sound	Buzzer transistor's base, through 1 kΩ

A pad printed GPn on the RP2040-Zero is the RP2040's GPIOn.

The chip's pull-ups on GP0, GP1 and GP2 add to the encoder module's own. The buzzer pin is set low first thing at start-up. A display transfer gives up after 25 ms. GP16 drives the board's RGB LED; the firmware leaves it alone. Every other pin is unused.

3 Build and flash

In `firmware/`, `pio run` builds `.pio/build/gerard/firmware.uf2`. Open and close the board's serial port at 1200 baud to restart it into its USB bootloader, then load the file:

```
pio run
picotool load -v -x .pio/build/gerard/firmware.uf2
```

picotool checks the write, prints "OK" and restarts the board. The build must run in a git clone of the repository: the version, `v.` + `firmware/VERSION` + the commit's UTC time, is read from git. A board with no saved settings starts with every default (Table 4).

> Note. With sound on, the boot tune plays after a flash. Switch sound off on the serial portal first if needed.

4 Use GERARD

Start from the menu. START begins a focus run, counting up. DOWN opens the countdown setting; a press starts it. After start-up and RESET it offers 30 min by default; otherwise it keeps the last value set. STOP ends the run. DOWN on the focus pause menu ends the run and opens the countdown setting. The break texts are “BREAK!”, “3 SIPS!”, “BREAK!”, “STRETCH!”, “BREAK!” and “20′ 20sec”.

4.1 Top line

- During a run: the run’s time in whole minutes, rounded down (“25”). A countdown shows the time run, not the time left. Each run starts at 0.
- On a pause menu: “PAUSED” and a clock, MM:SS or H:MM:SS: the run’s time, or the countdown’s time left.
- On the break prompt and “U GOOD?”: the run’s minutes, standing still.
- Elsewhere: the screen’s title (GERARD, COUNTDOWN, RESET SESSION, SLEEP) or nothing.
- With a storage fault, “! “ comes first (“! 25”, “! PAUSED 12:34”), or “!” alone (Section 6).

4.2 Screens

Table 2. Screens, and what the knob does on each. On SURE?/NO! screens, NO! is chosen first.

Screen	Shows	Turn	Press	Hold
Start-up	“GERARD”, version, commit; 2.5 s	menu	menu	menu
Menu	START, DOWN, RESET, POWER	next item	choose	nothing
Focus run	run view, or the time counting up	show the count again	view: show the count; count: pause	nothing
Focus paused	“PAUSED”, run time; RESUME, STOP, DOWN	next item	choose	resume
Countdown setting	minutes to count down (“30 MIN”), 1 to 240	±1 min a click; ±5 min turned fast (clicks under 60 ms apart)	start	menu
Countdown run	run view, or time left “OF 30 MIN”	show the count again	view: show the count; count: pause	nothing
Countdown paused	“PAUSED”, time left; RESUME, STOP	next item	choose	resume
Countdown done	“DONE!”, 60 s by default	menu	menu	menu
Break prompt	six break texts, 2 s each	resume the run	“U GOOD?”	“U GOOD?”
“U GOOD?”		resume the run	resume the run	resume the run
Reset	“RESET SESSION”; SURE?, NO!	other choice	choose	menu
“POOF!”	1 s, session cleared	menu	menu	menu
Power	“SLEEP”; SURE?, NO!	other choice	choose	menu
“SLEEP...”	1.5 s, then screen off	nothing	nothing	nothing
Sleeping	screen off	nothing	wake to the menu	nothing

4.3 Time, breaks and the session

Only running time counts: not a pause, the break prompt or “U GOOD?”. A break prompt comes after 30 min of running time (by default) since the last break prompt ended. A pause or a stop never counts as a break: the time toward a break adds up across pauses, stops and runs. Ending a break prompt starts it again; RESET clears it. The break tune plays when the prompt appears and every 60 s (by default) after, until it is answered. The prompt never dims and never turns off. A turn resumes the run; a press or hold shows “U GOOD?”, and any input there resumes the run and counts the break as answered.

The session’s focus time adds up every run since the last start-up, RESET or POWER; only the serial portal shows it. RESET clears it, the current run and the break tally. POWER clears it. Neither clears the lifetime stats.

4.4 Idle and dimming

The screen dims 2 min after the last turn or press, on every screen but the break prompt. A turn, a press, a break prompt and its reminders, or the countdown’s end bring it back. With no input, the screen shows “IDLE?” after 3 min, “IDLE?!” after 10 min and “IDLE?!?!” after 30 min, and turns off after 35 min. Each time is a default (Table 5), and each comes later than the one before. The first input after an idle prompt or that turn-off only wakes it. “U GOOD?” turns off after the screen-off time. The serial portal is not input.

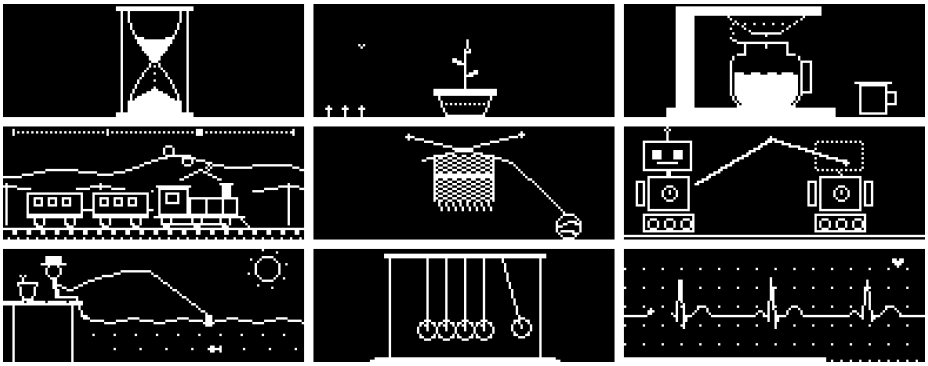
Dimmed: SSD1306 contrast 0x28, pre-charge 0x11, VCOMH 0x00. Full: 0xCF, 0xF1, 0x40. Contrast alone barely dims an SSD1306 while VCOMH is high.

4.5 Run views

During a run, an animation fills rows 16 to 63 at 10 frames a second: A hourglass, B plant, E coffee pot, F train, I scarf, J robot, K fishing, N Newton’s cradle, O heartbeat. A view is drawn at random among those switched on, at each run start and after each break prompt, never the one before. A paused run keeps its view. The count shows for 5 s (by default) at a start, a resume, or a turn or press, then the view slides in from the left over 0.3 s.

A view that shows progress fills over the break time during a focus run, so it is full when the break prompt comes; during a countdown, over its length (Figure 1). The plant grows a flower per break answered through “U GOOD?” this session, up to 7; RESET clears them.

Figure 1. The nine run views, about two thirds of the way to a break. Animated versions are in `docs/figures/`.



4.6 Tunes

Table 3. Tunes. Each is an RTTTL text of up to 200 printable characters, 64 notes and rests, and 10 s.

Tune	Plays	Default RTTTL text
boot	at start-up	`boot:d=16,o=6,b=250:g,c7,8e7`
wake	on waking from sleep	`wake:d=16,o=7,b=250:c,8e`
pause	at a pause	`pause:d=16,o=7,b=250:e,32p,8c`
resume	at a run's start and every resume	`resume:d=16,o=7,b=250:c,32p,8e`
break	at a break prompt and its reminders	`break:d=8,o=6,b=250:a,32p,a,32p,4d7`
done	at the countdown's end	`done:d=8,o=6,b=250:g,c7,e7,32p,c7,4.e7`
poof	at RESET	`poof:d=16,o=6,b=250:e7,c7,8g`
sleep	at POWER	`sleep:d=8,o=6,b=250:e7,c7,b,4g`

RTTTL: the Nokia ringtone text format.

5 Settings and the serial portal

Table 4. Settings.

Setting	Choices	Default
Sound	on, off	off
Knob direction	normal, reversed	normal
Display rows	standard (0x12), sequential (0x02): SSD1306 command 0xDA	standard
Large-line font	Inconsolata Bold, Logisoso, FreeUniversal Bold, Spleen	Inconsolata Bold
Top-line font	ProFont 15, Spleen 8x16, X11 7x14 Bold, FreeUniversal Bold 11	ProFont 15
Run views	each of the nine on or off; one at least stays on	all on

Table 5. Timings, set on the serial portal's [0] page. The idle times must each be later than the one before.

Key	Timing	Default	Unit	Range
a	Break every	30	min	10 to 90, steps of 5
c	Break tune repeats every	60	s	30 to 600, steps of 30
d	Screen dims after	2	min	1 to 30
e	"IDLE?" after	3	min	1 to 120
f	"IDLE?!" after	10	min	1 to 120
g	"IDLE?!?!" after	30	min	1 to 120
h	Screen off after	35	min	1 to 120
i	Count shows for	5	s	2 to 30
j	Countdown starts at, after start-up or RESET	30	min	1 to 240
k	Hold for "back"	700	ms	400 to 2000, steps of 100
l	"DONE!" shows for	60	s	10 to 300, steps of 10

A new timing is used at once and saved if it is in range, on its step and in order; otherwise the portal refuses it and says why. [r] puts every timing back to its default. The watchdog, the frame rate, the save interval, the short screens and the countdown's fast-turn step are fixed.

Settings and lifetime stats are saved every 5 min of unsaved running time, and at each pause, run end, break prompt, countdown end, RESET, POWER, start-up and setting change. Each save is read back before it replaces the old one. They live in a 64 KB area of the 2 MB flash, 0x101EF000 to 0x101FF000, which a firmware upload keeps. The timings have a file of their own, `/timings.bin`, saved the same way. With no file, or a damaged one, the defaults apply.

The lifetime stats count focus time, runs started, the longest run without a stop (up to 2^32 ms, about 49.7 days), countdowns finished, break prompts shown and answered, start-ups, and watchdog start-ups. Only portal key [6] clears them, after "yes" is typed; the start-up count survives.

The serial portal is on the board's USB serial port at 115200 baud. Its header shows the firmware version and commit. Each choice is one key; [b] or Esc goes back.

Table 6. Serial portal.

Key	Page
[1] Status	Screen, session focus time, current run, time since a break, sound, run view and breaks answered, brightness, timings, view drawing time, display found and its address, storage, saved record at start-up, last start-up, uptime
[2] Lifetime stats	The lifetime stats
[3] Sound	Sound on or off
[4] Buzzer and tunes	Play each tune, edit its RTTTL text, play a tone of a given frequency and length, play a sweep
[5] Hardware check	Board ID, I2C scan, live levels of the encoder lines, five display test patterns
[6] Erase lifetime stats	After “yes” is typed
[7] Restart the board	After “y” is typed
[8] Settings	Knob direction, display rows, fonts
[9] Run views	Keys 1 to 9 switch a view; 0 switches all on
[0] Timings	Each timing with its value and range; its letter sets it (Table 5)

> NOTICE. Never open the serial port at 1200 baud for the portal. Closing it at 1200 baud restarts the board into its bootloader, and GERARD stops until the firmware is started again.

6 Faults

Table 7. Faults and how GERARD shows them. No known problem is open.

Fault	How it shows
Storage not mounted, or a save failed	“! “ on the top line; Status shows `NOT MOUNTED` or `FAILED`
Saved record damaged at start-up	Lifetime stats restart from zero; their page says so
Display missing	Status: `NOT FOUND (looked for at 0x3C and 0x3D)`
Main loop stopped for 4 s	The watchdog restarts the board; Status shows `WATCHDOG RESET`
Saved tune unreadable	The built-in tune plays; the tunes page says “(saved tune unreadable: built-in used)”
Saved timings damaged	The defaults apply; Status and the Timings page say `DAMAGED`. No “!” on screen
Encoder unplugged	Nothing on screen: the lines read like a knob at rest. The Hardware check live page shows them
Button stuck down	One hold, then nothing until released. Held at start-up: no event until released

The firmware looks for the display at I2C address 0x3C, then 0x3D; GERARD's answers at 0x3C. A view frame takes about 4 ms to draw (4.3 ms at most seen) of its 100 ms. The portal's I2C scan takes 0.6 s at most, inside the watchdog's 4 s. Every "since" time is capped at 2^30 ms, about 12.4 days, so the millisecond clock's wrap at about 49.7 days shows nothing false.

7 Sources

- ThingPulse esp8266-oled-ssd1306, issue 134: with VCOMH at 0x40, contrast alone barely dims an SSD1306. https://github.com/ThingPulse/esp8266-oled-ssd1306/issues/134
- U8g2 2.36.18, `u8x8_d_ssd1306_128x64_noname.c`: the full-brightness values 0xCF, 0xF1, 0x40, and its note that VCOMH 0 does not suit every module.

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